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Roll No

AU-702(A)-CBGS

B.Tech., VII Semester

Examination, December 2020

Choice Based Grading System (CBGS)

Combustion and Heat Transfer

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. a) Compare SI and CI Engine with respect to Compression ratio and Ignition. 7
b) Why a rich mixture is required for maximum power? 7
2. a) What is supercharging in a IC engine? 7
b) What are the types of Combustion Chamber used in C.I Engines? 7
3. a) Explain ignition lag in S.I Engine. Discuss The effect of engine variables on ignition lag. 7
b) Explain the stages of combustion in a S.I. Engine. 7
4. a) Write short notes on diesel knock and its control. 7
b) Explain the Fouling of heat exchanger? 7
5. a) Illustrate the mode of mass transfer. 7
b) What is the difference between condenser and evaporator. 7

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6. a) Illustrate the general heat conduction equation of Cartesian coordinates for a rectangular body. 7
b) Explain the effect of the thermal conductivity when temperature is increases in case of metals and non metals. 7
7. a) What do you mean by critical thickness of insulation? Give their general equation for hallow cylinder. 7
b) What do you mean by overall heat transfer coefficient? Derive the general equation with suitable sketch. 7
8. a) How diesel knock differs from petrol engines knock? 7
b) What happens on the tendency of knocking in SI engines if engine speed increases. 7
